25900232

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU–UGP (HONOURS) Regular Examination – October 2025
Third Semester
Discipline Specific Core Course (DSC): MG3DSCBCH202 – TECHNIQUES IN BIOCHEMISTRY AND FORENSIC SCIENCE
(2024 Admission Onwards)

Duration: 1 Hour 30 Minutes **Maximum Marks:** 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A – Multiple Choice Questions
Answer all questions
Each question carries **1 mark**

1. The purpose of hybridization in Northern blotting is

   ANS: c) To bind a complementary probe to the RNA of interest **[U] / [CO1]**
2. The separation in gel filtration chromatography is based on _____.
   ANS: b) Size **[U] / [CO1]**
3. Dark field microscopy is used in
   ANS: b) Metallurgic Microscope **[U] / [CO2]**
4. Choose the person who coined the term "cell".
   ANS: a) Robert Hooke **[K] / [CO2]**
5. The type of DNA found in hair shaft is
   ANS: b) Mitochondrial DNA **[U] / [CO3]**

[1×5 = 5 marks]

Part B – Fill in the Blank Questions

Answer all questions
Each question carries **2 marks**

6. SDS is used as _____ in polyacrylamide gel electrophoresis.
   ANS: a denaturing agent (detergent) **[K] / [CO1]**
7. In affinity chromatography, the target molecule binds specifically to the _____.
   ANS: Ligand **[K] / [CO1]**
8. _____ is the part of a centrifuge that contains tube holders and drive shaft.
   ANS: Rotor **[U] / [CO2]**
9. The first electron microscope was constructed by _____.
   ANS: Ernst Ruska **[K] / [CO2]**

10. In maternity testing, the child's DNA is compared with that of the _____ mother.
    ANS: Alleged **[U] / [CO3]**

[2×5 = 10 marks]

Part C – Short Answer Questions

Answer any 5 questions

Each question carries **3 marks**

11. Explain why two dimensional paper chromatography is more useful than one dimensional chromatography in separating amino acids.
    ANS: Explain briefly two dimensional paper chromatography and One two dimensional paper chromatography - 2marks
    Advantage of two dimensional paper chromatography - 1Mark **[U] / [CO1]**
12. Discuss the principle of Southern blotting.
    ANS: Principle : 3 marks **[U] / [CO1]**
13. Explain the advantages of spectrophotometer over colorimeter.
    ANS: Any three advantages of Spectrophotometer over Colorimeter – 3 marks **[U] / [CO2]**
14. List any three major types of spectroscopy.
    ANS: Write any three major types including UV Visible Spectroscopy, IR Spectroscopy, NMR SPectroscpyetc- 3 marks **[K] / [CO2]**
15. Differentiate between direct ELISA and sandwich ELISA.
    ANS:define Elisa- 1.5 Marks
    Difference : 1.5marks **[U] / [CO3]**
16. Discuss the advantages and limitations of RIA.
    ANS: Advantages of RIA 1.5 marks
    Limitations of RIA 1.5 marks **[U] / [CO3]**
17. Recognize the function of the primer in PCR and give the typical temperature for the annealing step in PCR.
    ANS:: Function of Primer – 2 marks
    Temperature for annealing step in PCR- 50-65^{0}C **[K] / [CO3]**

[3×5 = 15 marks]

Part D – Short Essay Questions

Answer any 4 questions

Each question carries **5 marks**

18. Define Western blotting and recognize its significance in protein analysis.
    ANS: Explain Western blotting – 4 marks
    Significance of western blotting -1marks **[K] / [CO1]**
19. Describe the procedure and applications of TLC.
    ANS: Chromatography Definition - 0.5 marks
    Explain Principle and Procedure of TLC-3.5 marks
    Applications of TLC - 1 marks **[U] / [CO1]**
20. Illustrate the significance of ultracentrifuges in biochemistry and molecular biology studies.
    ANS: Principleof centrifugation-1 marks
    Explanation of Ultracentrifuge – 2 marks

Significance -2 marks **[A] / [CO2]**

21. Explain Beer Lambert's Law and identify the various components present in a colorimeter.

    ANS: Beer Lamberts law – 2 marks

    Explain the components of colorimeter with the help of diagram- 3 marks

    [U] / [CO2]

22. Discuss the various steps involved in DNA fingerprinting for maternity testing.

    ANS: Principle: 1 marks

    Steps in DNA Finger printing – 4 marks **[U] / [CO3]**

23. Discuss how skin cells, urine, faeces and teeth became valuable sources of DNA in forensic investigations.

    ANS: Explain valuable sources of DNA in forensic investigations

    Skin cells – 1.5 marks

    Urine - 1marks

    Faeces-1marks

    Teeth – 1.5 marks **[U] / [CO3]**

[5×4 = 20 marks]